

Scalar values in downward and non-downward entailing contexts

M.A.C. Huijbregts
ILS/UU/Febr 26, 2026

ABSTRACT. We discuss (i) the nature of a lexicalization condition on scalar implicatures that precludes lexicalizing negation of strong scalar elements, and (ii) parametric variation in the scope properties of conjunctive and disjunctive operators relative to negation. The empirical cases discussed show that logic differs from natural language, an internal cognitive system and part of human biology, therefore empirically constrained, in this case by conditions on polarity and scalar implicatures that are absent from formal logic. The interpretation of disjunction and conjunction is universal as in propositional logic following de Morgan's laws. Hence, the conjunctive interpretation of disjunction in negative statements comes naturally. Still, how these laws apply in propositional logic and how humans attain knowledge of these laws and their use in language are different empirical problems for a biological organism to solve. It's an intriguing empirical question which obviously needs an answer. Guided by the subset principle, children at first adopt the scope interpretation that yields true sentences in the narrowest range of circumstances, only "unlearning" what comes natural to them when forced to do so by positive evidence only. Biologically driven human capacities such as innate logic and the faculty of language, interacting with general third-factor principles of learnability and computational efficiency, guarantee that not everything need be learned and what need be learned can be learned, providing a principled solution to the poverty of stimulus for the acquisition of language.

Keywords Scalar implicature, strong/weak scalar elements, valuation space, resource restriction, subset principle, conjunctive/disjunctive parameter, de Morgan's laws, scope interpretation

1. Universal constraints on conceptualization. Human language is more than just merge-based syntax constrained by general third factor principles of simplicity, computational efficiency, and resource restrictions that apply generally beyond language. In addition, syntax must draw on a store of "lexical items," concepts that are universally shared and must satisfy general (conceptual, empirical) limiting conditions. We understand Escher's impossible drawings of infinitely descending and ascending stairs or his Penrose triangle as illusions, objects that cannot exist in reality. A married bachelor or a square circle is logically excluded.¹ A predicate "convince X of Y" without entailment that X believes Y cannot be a lexical concept.

There are also concepts that are not ruled out *a priori* but do not exist anyway. A few cases are discussed in some detail below. We have words for the colors 'blue' and 'green' but not for non-projectible logical predicates 'grue' or 'bleen' (figuring in Goodman's new riddle of induction); for the natural kind 'rabbit' but not for an unnatural category 'undetached rabbit part' (figuring in Quine's indeterminacy of translation). Natural languages only allow conservative determiners like, e.g., *every* as in "every bear is brown," equivalent to "every bear is a brown bear," and exclude non-conservative determiners like, e.g., *yreve*, roughly equivalent, let's assume, to "only," except that the latter is not a determiner, so that we get "the bears include all brown things," which has different truth-values from "the bears include all brown bears" (von Fintel & Keenan, 2016; Knowlton et al., 2020). Language has activity verbs like *seek* ("try to find") but no verb *SEEK* ("try to find and do find"), denoting a successful search activity, so that in contrast to *We sought but didn't find the missing child*, the sentence *We SOUGHT but didn't find the missing child* would be a contradiction.

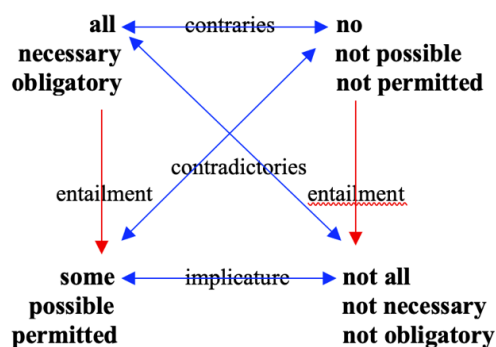
Mereological perception of how parts relate to the whole must be an important capacity, essential to understanding the world. Here as well, semantic constraints on a system of humanly accessible concepts may severely limit the choices that can be made. To give an example, some languages have collective nouns. E.g., the Dutch nouns *ge-boom-te* (“collection of trees”), *ge-blader-te* (“foliage”), *ge-berg-te* (“mountain range”) constructed with the “collective” circumfix *ge...te*. The collective of leaves refers to the leaves on the branches of trees, not the scattered leaves fallen on the ground. The non-existence in human language, apparently, of a collective noun for the latter follows from deep-seated semantic principles shared by all humans. Similarly, *ge-been-te* (“skeleton, carcass”) refers to some anatomically structured collection of bones, not to some collective of *legs* even though *been* is ambiguous between “leg” and “bone” in Dutch. Similarly, we speak of a swirling and shape-shifting murmur of starlings, a spatiotemporally contiguous but scattered object in motion. Paraphrasing Chomsky, no natural language, apparently, has a word “FLOCK” like the collective noun “flock” but denoting a single scattered object with starlings as parts, so that “a starling broke a wing” entails “the FLOCK broke a wing” (Chomsky, 1965).

2. Modality, quantification, and conjunction. There are more systematic gaps that are not excluded a priori. What causes the contrast in Dutch between weak scalar terms *iets/niets* [“something/nothing”] and strong scalar terms *alles/*nalles* [“everything/not-everything”]? Why is the adjective *innécessaire* an impossible word, not only in French, but in all languages despite English *unnecessary*? We can *disturb* the peace. Why can’t we *undisturb* the peace that has been disturbed? After all, peace may be *undisturbed*. Here *undisturb* instantiates a systematic lexical gap that is explained by an empirical condition that holds generally, namely Subjacency: No two “negative” prefixes may be subjacent. The existence of the adjectives *undisturbed* and *ununlockable* (“can’t be unlocked”) can now be explained. The “negative” prefixes show linear adjacency but are structurally separated by the intervening affixes *-ed* and *-able* in their respective hierarchical lexical structures, $\{un-, \{-ed, \{dis-turb\}\}\}$ and $\{un-, \{-able, \{un-, lock\}\}\}$. The two linearly adjacent prefixes are structurally non-subjacent in their merge-based structures or their planarized phrase structures $[un- [[dis-turb] -ed]]$ and $[un- [[un-lock] -able]]$, thus explaining the constrained ambiguity. The latter can only mean “unable to be unlocked,” while the simpler *unlockable* is ambiguous between $\{un-, \{-able, lock\}\}$, i.e., “unable be locked,” and $\{-able, \{un-, lock\}\}$, i.e., “able to be unlocked.” Double negatives are lexically excluded. Notice that while “double negatives” are excluded from lexical concepts, they are licensed in syntax as is illustrated in e.g., *John could not have not been a victim of his upbringing* (“he definitely must have been a victim of it”), *John couldn’t not have accepted the invitation* (“he had been obliged to accept it”), or *John can’t not admire her* (“he can’t but admire her”).

The rich variety of languages resides in their lexicons and morpho-phonologies. But that doesn’t mean that lexical differences code for different conceptualizations of the real world. In fact, lexical processes in the world’s languages must meet general constraints that set strong limits on conceptualization. Apparently, although there’s no a priori reason for it, no language has an affix denoting logical necessity rather than contingency. E.g., the Turkish abilitative suffix *-ebil/-abil*, denotes logical possibility in *öl-dür-ebil-ler* (“they could/might kill”), while its necessitative counterpart *-malı/-meli* denotes evidentiality, certainty, or obligation as in *öl-dür-meli-ler* (“they must/should kill”) but not logical necessity. Similarly, the modal adverbs *maybe*, *peut-être* or *olabilir* contrast with their necessitative counterparts **must-be*, **doit-être* or **olmalı*. See Larry Horn (1972, 1989) for substantial discussion. Even the adjective “unnecessary” itself must be deontic in English as in other languages

generally rather than express negation of logical necessity: **innécessaire* (French), **onnoodzakelijk* (Dutch), **onödvändig* (Swedish), **olmaluma* (Turkish) do not exist in the respective languages. Still, these languages do contain their deontic counterparts *inutile*, *onnodig*, *onödig*, and *gereksiz* expressing lack of obligation. Japanese contains *hitsuyō* (“necessary”) and *fu-hitsuyō* analogous to *kanō* (“possible”) and *fu-kanō* (“impossible”) but *fu-hitsuyō* can be understood only as a synonym of “non-obligatory,” i.e., “permitted-not” rather than “possible-not,” precisely as English “unnecessary” (Hisatsugu Kitahara, pc).

2.1. Quantification and modality: existential possibility & universal necessity. There’s no a priori reason for any of this to be the case, but the lexicalization condition has strong empirical support. Negative contraction is universally blocked for all strong scalar lexical items. The generalization that “ $\neg \Box$ ” (not logically necessary) does not lexicalize in contrast to “ $\neg \Diamond$ ” (not logically possible) must be extended to domains of quantification and conjunction. That is, $\Box$, $\forall$, $\wedge$, involving strong scalar values, contrast with $\Diamond$, $\exists$, $\vee$, involving weak scalar values. Compare Dutch *iets/niets* vs. *alles/*nalles* or English *or/nor* vs. *and/*nand* (Horn, 1972; Huybregts, 1979; Jaspers, 2005). Therefore, there must be a profound universal principle explaining why logically possible lexicalizations are empirically excluded from human conceptual systems: **innecessary/impossible*, **nall/none*, **nalways/never*, **nand/nor*. Empirically, logical negation of necessity or conjunction is a systematic gap (“a blind spot”) in humanly accessible lexical concepts. The lexicalization constraint applies more generally to all strong scalar values of the quantificational, modal and logical operator systems. Neither lexical nor morphological rules access logical necessity, universal quantification, or logical conjunction. Horn (1972, pp 263-4) suggests a lexicalization condition on scalar implicatures, starting with the observation that scalar predicates constitute pairs of stronger/weaker scales, $\langle \text{necessary}, \text{possible} \rangle$, $\langle \text{all}, \text{some} \rangle$, $\langle \text{always}, \text{sometimes} \rangle$, $\langle \text{and}, \text{or} \rangle$. If a stronger scalar predicate S entails a weaker scalar predicate W, then W implicates $W \sim \equiv \sim S$, and $W \sim \equiv \sim S$ needn’t lexicalize, hence doesn’t lexicalize for reasons of (lexical) economy. A conversational implicature carried by W cannot be lexicalized. For example, the use of *some* carries the implicature that the stronger scalar predicate *all* does not apply, i.e., *some* implicates “some-not” $\equiv$ “not-all” and therefore “not-all” does not lexicalize as **nall*.



Strong scalar “all” entails weak scalar “some”
and “some” implicates “some-not” $\equiv$ “not-all.”
The implicature does not lexicalize (**nall*).

While formal logic is not biologically constrained and accepts “not-and” operators (e.g., NAND gate in digital electronics) or “not-all” quantifiers, natural language is an organic system with strong empirical restrictions imposed on it. Besides the pragmatic blocking condition on lexicalizing scalar implicature, there is another account that involves logical presupposition. Assuming a non-emptiness presupposition for universal quantification in natural language, preservation of presupposition may be a condition on lexicalization processes (Jaspers, 2005, pp 219-227).² The natural logic explanation that Jaspers advances adopts the biolinguistics approach of I-language studies in generative minimalist grammar. Generally, if a scalar predicate S entails an

existential presupposition P, then $\sim S$ cannot be lexicalized. It follows that, e.g., **n-and*, **n-all*, **n-always*, **innecessary* are simply excluded from being accessible lexical concepts of human language. More precisely, in natural language, A-operators/stronger scalar predicates (universal quantifier, conjunction, modal logic necessity) entail the meanings of the corresponding E-operators/weaker scalar predicates (existential quantifiers, disjunction, modal logic possibility) as presuppositions. Since the element “n-” in an A-operator (e.g., “n-all” or “n-and”) cannot cancel the non-emptiness presupposition carried by the operator, it contradicts that presupposition. But preservation of the presupposition is required for lexicalization. Hence, lexical inaccessibility is explained as a violation of the condition.

There may be another and independent alternative account based on computational efficiency. Adopting a principled I-language perspective that takes resource restrictions seriously for an organic system, we propose that lexical incorporation of negation in (strong) scalar predicates may not downsize falsification space. The valuation spaces of the truth-conditional value T or 1, differ for AND, OR, NAND, and NOR in interesting ways. The truth-functional conditions for OR and NOR are disjoint like those for AND and NAND, but while the set of circumstances that verify a positive statement for disjunctive OR is greater than the one for negative disjunction N-OR, i.e. $|\text{OR}(1)| > |\text{N-OR}(1)|$, the set of circumstances that verify a positive statement for conjunctive AND is smaller than the one for negative conjunction N-AND, i.e., $|\text{AND}(1)| < |\text{N-AND}(1)|$. In order to find a circumstance that would falsify $P \& G$, we need to find a single “false” term (three chances out of four), while we need to find two “true” terms to find a circumstance that would falsify $P \text{ and } Q$ (one chance out of four). Higher probability for falsification should be a computational desideratum. Therefore, if we assume that lexicalization must optimize falsification, i.e., increase the valuation space for the value F, lexicalization of “not-OR” (and, analogously, of “not-SOME” and “not-POSSIBLE”) will be licensed, while lexicalization of “not-AND” (and, analogously, of “not-ALL” or “not-MUST”) will be excluded as a violation of a computational resource restriction.

Why <i>nor</i> (<i>nothing</i>) and not <i>*nand</i> (<i>*nall</i>)?					
P	Q	P & Q and	not (P & Q) <i>*nand</i>	P ∨ Q or	not (P ∨ Q) nor
1	1	1	0	1	0
1	0	0	1	1	0
0	1	0	1	1	0
0	0	0	1	0	1

Lexicalization condition:

$$|\text{not-X}(1)| < |X(1)|$$

Lexicalization of **nand* decreases falsifiability (.25), while lexicalization of *nor* increases falsifiability (.75)

OR (= *either* or *both*) $\Rightarrow$ NOR (= *neither*)

AND (= *both*) $\Rightarrow$ NAND (= *either* or *none*)

To seek falsification for AND(1,0,0,0), NAND(0,1,1,1), OR(1,1,1,0) and NOR(0,0,0,1), one needs to find, resp., one value 0 (in 3 out of 4 situations), two values 1 (in 1 out of 4 situations), two values 0 (in 1 out of 4 situations), one value 1 (in 3 out of 4 situations). Since NOR is easier to falsify than OR, lexicalizing *nor* from “not-or” is unproblematic. Conversely, NAND is harder to falsify than AND, hence computational efficiency blocks lexicalizing **nand* from “not-and.” Upsizing falsification space (increasing probability of finding a falsification, thus making falsification easier) is equivalent to downsizing verification space (decreasing probability of finding a verification, thus making verification harder). Since falsification is easier than verification in the case of *nor* relative to *or*, it makes sense, therefore, to frame the L(exical) Re(source) R(estriction) on negative contraction in terms of a restriction on resources, namely ease of falsification.

LRR: Neg-contraction only applies if it facilitates falsification by downsizing verification space.

Verification searches would give contrary results. Finding a *single* value T for *nor* (probability .75) would lead to easy falsification, while verification requires *two* values F (probability .25). Lexicalization of **nand*, however, would require two values T for falsification, making a search more difficult rather than easier.³

To summarize this section, we discussed three proposals ruling out lexical incorporation of negatives in scalar predicates formulated in terms of pragmatics, natural logic, and computational efficiency. The different factors involved — no lexicalization of $\sim S$ if (a) it is the scalar implicature of a weaker scalar predicate W , if (b) the existential presupposition entailed by the stronger scalar predicate S is not preserved, or if (c) resource restrictions ruling out the downsizing of falsification space are violated — may be limiting factors for a learning organic system, and probably necessary for such a system to function at all.⁴

2.2. Learnability and the subset principle. The set of circumstances that verify a positive statement *A and B* is a subset of the circumstances that verify *A or B*. While the T-values of the truth-table of OR (1,1,1,0) include those of AND (1,0,0,0), the T-values of the truth-table of NAND (0,1,1,1) include those of NOR (0,0,0,1). See Figure X above. These subset/superset relations are significant factors in the explanation of some intriguing aspects of language development in the child. Children speaking Mandarin Chinese initially apply *de Morgan's Law* reflexively to the equivalent of, e.g., English “I didn’t drink any wine **or** beer” (i.e., *neither*) equivalent to “I didn’t drink any wine **and** didn’t drink any beer” (i.e., *both-not*). Later, upon receiving positive evidence from their parents, they subsequently come to understand the expression “disjunctively” as *not-both* (“I didn’t drink any wine **or** didn’t drink any beer”), differently from English (Crain, 2008; Crain & Thornton, 2012). These cross-linguistic differences in scope relations allow precise and specific predictions about the course of language development.

- (1) Ted didn’t order sushi or pasta (both-not, (45) of Crain & Thornton, 2012)
“It’s **not** sushi **or** pasta that Ted ordered”

The negative disjunctive statement generates a conjunctive entailment “Ted didn’t order sushi and Ted didn’t order pasta” in English (1) but not, e.g., in Mandarin (2).

- (2) (Wo cai) Ted **meiyou** dian yidalimianshi **huozhe** shousi (not-both, (46) of Crain & Thornton, 2012)
(I guess) Ted **not** order pasta **or** sushi
“It’s either pasta **or** sushi that Ted did **not** order”

In Mandarin, the disjunction word *huozhe* scopes over negation; in English, negation takes scope over *or*. It will be shown below that despite these differences in the scope relations between negation and disjunction, the meaning of disjunction, namely inclusive-*or*, remains fixed across human languages and conforms to de Morgan’s law. The languages only differ in the parameter values of a disjunction parameter. Mandarin adopts the “plus” value of the disjunction parameter (i.e., disjunction is a positive polarity item, **OR** = +PPI), while English adopts the “minus” value (i.e., disjunction is not a positive polarity item, **OR** = –PPI). When the parameter value

is “minus,” negated disjunctions generate conjunctive entailments. When the parameter value is “plus,” negated disjunctions generate ‘disjunctive’ truth conditions. Negated disjunctions yield disjunctive “**not-both**” readings ($\sim P \vee \sim Q$) when the disjunction is a positive polarity item, as in Mandarin, but generate conjunctive “**both-not**” entailments ($\sim P \& \sim Q$) when the disjunction is not a positive polarity item, as in English (Crain 2008; Crain & Thornton, 2012).

Similarly, the conjunction word *he* is a positive polarity item in Mandarin, AND = +PPI, where negated conjunctions generate the ‘**both not**’ (*neither*) readings, but not in English, where *and* is not a positive polarity item, AND = -PPI, and negated conjunctions generate the ‘**not both**’ readings (Crain 2008; Crain & Thornton, 2012). Again, the Mandarin word for conjunction *he* and the English word *and* have the same meaning but differ in the values of a conjunction parameter. The difference again is a difference in scope relations between negation and conjunction as illustrated in (3) and (4).

- (3) Ted didn’t order (both) pasta and sushi (not-both, (47) of Crain & Thornton, 2012)
 “It’s **not** both pasta **and** sushi that Ted ordered”

- (4) Taide **meiyou** dian yidalimianshi **he** shousi (both-not, (48) of Crain & Thornton, 2012)
 Ted **not** order pasta **and** sushi
 “As for both pasta **and** sushi, Ted did **not** order them”

Adult speakers of these languages interpret disjunction and conjunction as having scope over negation (OR>NEG, AND>NEG). It would then follow that de Morgan’s laws do not apply. Apparently, adult speakers “violate” de Morgan’s laws in simple negative sentences. But, as expected, appearance is deceptive. First, adult speakers of Mandarin or Japanese, and all other languages including English, adhere to de Morgan’s laws when disjunction is in a lower clause than negation. Disjunction licenses conjunctive entailments in the scope of nonlocal negation as in (5), which means “Mary didn’t say that John can speak French **and** Mary didn’t say that John can speak Spanish.” Sentences (5) and (6) are taken from Crain (2008).

- (5) *Mali meiyou shuo-guo [s Yuehan hui shuo fayu huozhe xibanyayu]*
 Mary no say John can speak French or Spanish
 “Mary didn’t say that John can speak French **or** Spanish”

Second, disjunction licenses conjunctive entailments in the scope of downward entailing expressions across all languages as in (6), which means “Every student who speaks French is in this room **and** every student who speaks Spanish is in this room.”

- (6) *Meige [s hui shuo fayu huozhe xibanyayu de] xuesheng dou tongguo-le kaoshi*
 Every can speak French or Spanish DE student DOU pass-Perf exam
 “Every student who speaks French **or** Spanish passed the exam”

These empirical results show that the *different* truth-conditions associated with disjunction in downward and non-downward entailing contexts are just different interpretations of the *same* lexical items that conform to de

Morgan's laws in propositional logic. Under non-local negation and in downward-entailing contexts, inclusive-*or* has conjunctive entailments satisfying de Morgan's law across all languages. In contrast, under local negation, languages show parametric variation: disjunction licenses conjunctive entailment in English, Dutch, and the like vs. disjunctive truth-conditions in Mandarin, Russian, and other languages. Overall, disjunction and conjunction words have invariant meanings in human language as inclusive-*or* and conjunctive-*and*, respectively, and only differ in parametric values ($\pm$ PPI) that are responsible for generating different entailments and truth-conditions in downward-entailing and non-downward entailing contexts. Finally, in all languages, inclusive-*or* invites scalar implicatures that strengthen inclusive-*or* to exclusive-*or* readings in non-downward entailing contexts.

Interestingly, children developing Mandarin will initially interpret (2) as similar to English (1). Conversely, children developing English will initially interpret (3) as similar to (4). For instance, child speakers of Mandarin, Japanese, Russian, Hungarian, and other languages interpret sentences like (2) as disjunction under negation (NEG>OR) so that they generate a conjunctive entailment as in de Morgan's law: $\text{not}(A \vee B) \Rightarrow \text{not-}A \ \& \ \text{not-}B$. The result is that the Mandarin word for disjunction, *huozhe*, does not have a different meaning than English *or*. The difference between children and adults is that children adopt a more restrictive (subset) value of a parameter that governs the scopal interpretation of disjunction under local negation.

The parameter settings of OR = +PPI (Mandarin) and AND = -PPI (English) are therefore not the initial parameter values. The initial settings are OR = -PPI and AND = +PPI. A result of an important learnability condition that children learn from positive evidence only (Berwick, 1985; Yang, 2016). Language growth in the developing child initially favors the scope interpretation that yields a true sentence S in the narrowest range of circumstances. In accord with the Subset Principle (Berwick, 1985), children adopt the subset value and change to the superset value only when forced to do so by positive evidence only. This happens in the case of Mandarin for disjunction, and in the case of English for conjunction. The child is presented with primary linguistic data that contain (negated) conjunctions as well as (negated) disjunctions. The subset principle then determines the smallest range of situations for which a simple sentence like *John didn't drink or smoke* is true relative to a scope interpretation NEG>OR vs OR>NEG, working out a solution simply from the truth-conditions of innate logic rather than from planarized sentences containing these logical operators. The circumstances for which the scope interpretation Neg(P-or-Q) makes a sentence S true, here **(0,0)** is a proper subset of the circumstances for which the scope interpretation **not-P or not-Q** makes S true, here **{(0,0), {0,1), (1,0)}**. The notation “(0,1)” used here represents the pair of truth-values False and True for P and Q respectively.

OR = $\pm$ PPI		AND = $\pm$ PPI		
child	adult	child	adult	
--	--	+	+	Dutch
--	--	+	--	English
--	+	+	+	Russian
--		+	--	unknown

Mandarin, Japanese, and Hungarian are like Russian, and represent languages from widely different language families: Sino-Tibetan, Altaic, Uralic, Indo-European.

Since the subset value of “or” is “minus” for PPI, Mandarin children start the same way English children do but reset the parameter to superset value when hearing positive evidence from the adult language, changing from the ‘neither’ to the ‘not-both’ interpretation. Likewise, since the subset value of “and” is “plus” for PPI, English

children start the same way Mandarin children do but reset the parameter to superset value when hearing positive evidence from the adult language, changing from the ‘both-not’ (*neither*) to the ‘not-both’ reading. In both cases, the set of ordered pairs of truth values of “P” and “Q” for which a propositional reading “both-not” is true is a subset of the set of ordered pairs of truth values of “P” and “Q” for which a propositional reading “not-both” is true.

To complete the picture, Dutch *Jan eet en drinkt niet* (John eat.3sg and drink.3sg not) has the SAME propositional meaning as *Jan eet of drinkt niet* (John eat.3sg or drink.3sg not), i.e., the **both-not** (*neither*) reading, and is preferentially used. In this respect Dutch, like Mandarin but unlike English, retains the initial parameter setting [+PPI] for conjunctive “and” and, like English but unlike Mandarin, retains the initial setting [-PPI] for disjunctive “or.” In contrast, English and Mandarin change their initial parameter settings for conjunctive “and” and disjunctive “or” respectively, and do so, in each case, from the *subset* to the *superset* of the truth-conditional values for which the expression is true. The initial states of Mandarin and English children are preserved as the steady states of Dutch adults, OR = -PPI and AND = +PPI, and explain why negated disjunction and negated conjunction generate identical entailments (‘both-not’) in Dutch. Finally, a fourth kind of language is predicted, which starts out with initial settings (as in Dutch) but later changes parameter settings of both “or” (as in Mandarin) and “and” (as in English) on the basis of positive evidence. Presently, the existence of reverse-Dutch, with OR = +PPI and AND = -PPI, is not (yet) attested. If this language type turns out not to exist, it raises the question if its non-existence is merely accidental or principled.⁵

The discussion is a useful illustration of how empirical constraints (subset principle, principles and parameters of UG) are limiting factors in what can be learned and needs to be learned. *Lexical* representation of “not-both” (**nand*) cannot be naturally learned since learning is excluded by third factor conditions, which, in contrast, do allow *syntactic* representation of “not-both” readings with the polarity values of adult Chinese “or” or adult English “and.” These values need to be learned and can be learned conforming to third factor principles. Language development in the child showing a change from conjunction-scoping-over-negation to negation-scoping-over-conjunction, reversing the course of development in Mandarin, Japanese, and other languages, would be excluded by the subset principle, a general, third factor resource restriction that is freely accessible to the developing organism.

3. Summary. Disruption of de Morgan’s laws is an illusion. Inclusive-*or* has the same meaning across languages. Logic is innate. The interpretation of disjunction is universally inclusive-*or* as in propositional logic following de Morgan’s law. Hence, the conjunctive interpretation of disjunction in negative statements comes naturally.

How de Morgan’s laws apply in propositional logic and how humans attain knowledge of these laws and their use in language, empirical problems for a biological organism to solve, are separate issues. The interpretation of disjunction seems to be part of the human genome. The empirical cases discussed here also show that logic differs from natural language (UG), an internal cognitive system that is unconscious and furthermore inaccessible to consciousness, but is part of human biology, and therefore empirically constrained, in this case by conditions on polarity and scalar implicatures that are absent from formal logic. Scalar implicature of exclusivity is a widely known phenomenon in language development and follows from a pragmatic preference for complete answers.

Children initially adopt the inclusive-*or* reading with hardly any relevant empirical data (Crain & Thornton, 2011). However, the lexical items for inclusive-*or* in different languages may have different settings for parametric values that yield empirical variation in specific contexts. Locally negated inclusive-*or* may be a positive polarity item with a “minus” value (narrow scope) generating conjunctive entailments when as in English, German, etc., or with a “plus” value (wide scope) generating disjunctive truth-conditions as in Mandarin, Russian, etc. Furthermore, in non-downward entailing contexts and in non-negative statements, adult use of inclusive-*or* is universally governed by a scalar implicature of exclusivity.

Why, then, would children sometimes “unlearn” what comes natural to them? That’s an intriguing empirical question which obviously needs an answer. There’s an immediate answer in terms of properties of learnability theory: the *Subset Principle* of Robert Berwick (1985), a third-factor principle applying to a biological system with properties that are uniquely linguistic and uniquely human. Children initially favor the scope interpretation that makes sentences true in the narrowest range of circumstances. In accord with the *Subset Principle*, children adopt the subset value and change to the superset value only when forced to do so by positive evidence only. Adult use of Mandarin disjunctive-*or* and English conjunctive-*and* is thus explained. The model predicts the potential existence of another type of language in which the initial values for conjunction (UG, Mandarin) and disjunction (UG, English) words remain fixed. This language will have a negative PPI value for *or* and a positive PPI value for *and*. This language is exemplified by Dutch. There’s a fourth logical possibility. Does a language exist that changes the initial values of both conjunction and disjunction parameters? This would be an English-type language with Russian values for *or*, or equivalently, a Russian-type language with English values for *and*.

Finally, minimal computation for maximal falsification would permit lexical *nor* while ruling out lexical **nand* across all languages. The principle functions as a resource restriction ruling out any lexicalization of scalar elements that would downsize the falsification space of their truth-conditions. It explains the lexicalization condition on scalar implicatures that precludes lexicalizing negation of strong scalar elements.

In all these cases, biologically driven human capacities such as innate logic and the faculty of language (UG), interacting with general third-factor principles of learnability and computational efficiency, guarantee that not everything need be learned and what needs to be learned can be learned. A perfect solution to the Poverty of Stimulus for the acquisition of human language.

_____ * * * _____

Appendix

Somewhat similar to Dutch or Mandarin, historically unrelated and geographically discontinuous Cheyenne (Algonquian) and Warlpiri (Pama-Nyunga) may be languages with positive polarity “and.” In Cheyenne the connective *naa* (“and”) must receive a disjunctive interpretation, but only in alternative Y/N questions (Murray, 2017).

- (1) Mó='-é-naā'e naa mó='-é-ma'heón-o'éétahe?
 Q=EP-3-doctor CONN Q=EP-3-sacred-do.something
 "Is he doctoring or is he a magician?"

The alternative question invites answers that include "he is a doctor" but exclude "yes" or "no." There are interrogative proclitics on both sentences that are conjoined by the connective *naa*, which, therefore, scopes over the interrogative elements: $\&(Q-A, Q-B)$. So how do we get from here to $Q(A \vee B)$, if at all?

The question may be analyzed as asking both whether or not he is a doctor and whether or not he is a magician. Possible answers to the conjunction scoping over questions, $\&(Q-A, Q-B)$, are listed in (a) through (d) and are equivalent to potential answers to the question scoping over disjunction.

- (2) [Tell me] Is (or isn't) he a doctor and is (or isn't) he a magician?
 a. He is a doctor and a magician ("he is both")
 b. He is a doctor (not a magician)
 c. He is a magician (not a doctor)
 d. He is not a doctor, and he is not a magician ("he's neither")

Discourse context determines the correct/felicitous response. A simple "yes" or "no" is not a felicitous answer but "he is a magician, not a doctor" may be. Another case mentioned by Murray is an alternative question raised in a context where "the speaker is asked to pass a cup, but he's not sure which cup was intended."

- (3) mó=hé'tóhe naa mó=hé'tóhe
 [Q=this.one "naa" Q=this.one
 "this one (pointing) or this one (pointing)?"

Again, four potential kinds of answers exist but discourse pragmatics dictates the exclusive answers "this one" or "that one," eliminating other options. Pragmatics of the conversational discourse reduces the conjunctive question to an exclusive disjunctive question. On this account, the problem seems to have vanished. In other contexts, 'compositional' connectives *naa mātō=héva* (CONN also=maybe) or *naa mó=héá'e* (CONN Q=maybe) can be alternatively used to express disjunction.

Warlpiri connective *manu* is different again. The connective is a conjunction in matrix positions but a disjunction in negative statements ("embedded positions"). The sentences below are from **Margit Bowler (2014)**. Conjunctive "X *manu* Y" in (4), i.e., "both X and Y," gets interpreted disjunctively in the scope of negative prefix *kula* in (5), i.e., not (X or Y).

- (4) Cecilia **manu** Gloria=**pala** yanu tawunu-kurra. Jirrama=juku.
 Cecilia "manu" Gloria =3DU.SUBJ go.PST town-ALL. two=exactly
 'Cecilia and Gloria went to town. Exactly two did.'
- (5) Cecilia **manu** Gloria **kula=pala** yanu Lajamanu-kurra. Lawa.
 Cecilia "manu" Gloria NEG=3DU.SUBJ go.PST Lajamanu-ALLL. Nothing
 'Neither Cecilia nor Gloria went to Lajamanu. Neither one did.'

Bowler (2014) argues for recycling “or” to express “and” in specific contexts. If A and B is true, A or B is also true, but not conversely. On her analysis of Warlpiri, the inclusive disjunction *manu* gets “recursively” strengthened to a conjunction (rather than exclusive disjunction) in non-negative contexts.

Strengthening a basic meaning by computing scalar implicatures in contexts that are not downward entailing is a common phenomenon in language development and follows from a pragmatic parsing preference for complete answers (Singh, Wexler, Astle, Kamawar & Fox, 2016). E.g., given *John drank beer or wine*, the scalar implicature “John didn’t drink both beer and wine” yields the exclusive disjunction that John drank beer or wine but didn’t do both: $(P \vee Q) \wedge \neg(P \wedge Q)$. Notice that there is no scalar implicature (SI) in downward-entailing contexts that would enable the strengthening of inclusive disjunction to exclusivity. Consider the sentence, *If John likes pizza or pasta, he’ll leave for Italy*, which entails that “If John likes pizza, he’ll leave for Italy, and if John likes pasta, he’ll leave for Italy.” No exclusive disjunction is entailed by SI. Further empirical support may be found in downward-entailing disjunctive comparatives, which license negative polarity items as in *John is richer than Bob has ever been* or Dutch *Jan moet harder werken dan iemand ooit heeft hoeven doen* (“John has to work harder than anyone ever had to”), and require overt expression of a negative particle in French (*Cela va plus vite qu’on ne pense* or *Il est plus riche qu’il ne l’a jamais été*). The comparative sentence *John is richer than Bob or Charles* entails that John is richer than both Bob and Charles, i.e., *John is richer than Bob*, and *John is richer than Charles*. For downward-entailing comparative operator $-ER$, $-ER(A \text{ or } B)$ entails $-ER(A)$ and $-ER(B)$. Scalar implicature does not arise in DE-contexts, hence the exclusive disjunction, “Either John is richer than Bob, or else John is richer than Charles,” is not entailed.

However, this situation is different from Warlpiri, which on Bowler’s analysis only has an unstrengthened disjunctive coordinator. The lexicon does not contain a conjunction word. Scalar alternatives for English $P \text{ or } Q$ would be the set $\{P, Q, (P \vee Q), (P \wedge Q)\}$. Warlpiri speakers cannot substitute for *manu* because there are no lexical alternatives. The set of alternatives for $P \text{ manu } Q$ available to Warlpiri speakers is thus a simple effect of the lexical items that are available to them, namely the set $\{P, Q, (P \vee Q)\}$. The scalar alternatives to “ $P \text{ manu } Q$ ” exclude the truth-functional $(P \wedge Q)$. However, the “basic” disjunctive meaning of connective *manu* can be recursively strengthened to a conjunctive meaning in “non-embedded” (i.e., non-negative) contexts. First, the strengthened interpretations of $\{P, Q\}$ generate the set $SI = \{(P \wedge \neg Q), (Q \wedge \neg P)\}$, containing elements that are mutually exclusive. The next step, the conjunction of prejacent $(P \vee Q)$ with the negated propositions of SI yields the final strengthened conjunctive interpretation $((P \vee Q) \wedge \neg(P \wedge \neg Q) \wedge \neg(\neg P \wedge Q)) = (P \wedge Q)$ in unembedded contexts. E.g., “John drank beer *manu* wine” gets strengthened to “John drank beer, wine, or both, and if he drank one beverage, he also drank the other, so he drank both.” However, as suggested above, there’s an alternative analysis that simply interprets *manu* as a conjunction that adopts the “plus” value [+PPI] of the conjunction parameter.

Endnotes

¹ Squaring the circle with compass and ruler was proven to be mathematically impossible, a result of the *Lindemann-Weierstrass* theorem.

² Jaspers (2005) adopts a Strawsonian approach where universal statements “All F are G” carry existential import, a set-theoretic non-emptiness presupposition, rather than material implication (as in Russellian logic).

³ Sandiway Fong suggests that the reason why **nand* is not lexicalized may be that *NAND* is not associative, while *AND* is. While *A and (B and C)* is identical to *(A and B) and C*, the proposition *A nand (B nand C)* doesn't equal *(A nand B) nand C*, e.g., if both A and B have a truth value different from C. But *NOR* is not associative either. While *A or (B or C)* is identical to *(A or B) or C*, the proposition *A nor (B nor C)* doesn't equal *(A nor B) nor C*, e.g., if A and B are both true and C is false. Therefore, this cannot explain why “not-AND” does not lexicalize (**nand*), while “not-OR” does (*nor*). It's true that contrasting with propositional logic, the negated disjunction *nor* in natural language can be used associatively as in *John nor Bob nor Erwin like spinach*, i.e., true if none of them likes spinach. But this doesn't help because now the question arises why there wouldn't be an associative negative conjunction *nand* for cases like **John nand Bob nand Erwin like spinach*, i.e., true if at least one of them likes spinach. The answer that we happen to have *nor* but not *nand* begs the question. We would be back at where we started.

⁴ Huybregts (1979) stipulates a “blocking” condition on lexicalization. Lexicalization of “ $\sim\exists \equiv \forall\sim$ ” yielding attested “none” blocks lexicalization of “ $\exists\sim \equiv \sim\forall$ ” yielding excluded “nall.” Application of blocking presupposed the availability of lexical incorporation of negation. Hence, in languages like Hindi or Japanese without single word lexicalizations for negative scalar elements, the condition does not apply. Still, the account is merely descriptive since it leaves unanswered why lexicalization could not, in any language, have a different orientation with lexically well-formed “nall” and unattested “none.” Barwise & Cooper (1981) advance yet another blocking proposal that fails as an explanation for comparable reasons. Slightly adapting their *monotonicity correspondence universal* (p.186), there's a basic determiner D which expresses the monotone *decreasing* quantifier $\sim Q$ if and only if there's a weak non-cardinal basic determiner D' which expresses the monotone *increasing* quantifier Q. Thus, monotone decreasing “nall” would be excluded since monotone increasing “all” is a strong non-cardinal determiner: *all/*nall men* vs. *some/no men*. The matter may receive an explanatory answer, discussed in the main text, asserting that falsification space must not be downsized.

⁵ Suppose someone were to counter-argue that a right-node-raising analysis of (1) or a left-node-raising analysis of (2) would account for the equivalence of these cases with (1') or (2'), i.e., $\neg P \& \neg Q \equiv \neg(P \text{ or } Q)$.

- | | |
|---|-----------------------------------|
| (1) Jan [rookt niet] en [drinkt niet] | (1') Jan rookt of drinkt niet |
| (2) dat Jan [niet rookt] en [niet drinkt] | (2') dat Jan niet rookt of drinkt |

But then why wouldn't (1')-(2') themselves receive RNR/LNR analyses (3)-(4)? Why wouldn't one of de Morgan's laws, $\neg P \text{ or } \neg Q \equiv \neg(P \& Q)$, make these equivalent to (3')-(4'), i.e., John smokes or drinks but doesn't do both? However, each of (3')-(4') makes the stronger assertion that John neither smokes nor drinks.

- | | |
|---|-----------------------------------|
| (3) Jan [rookt niet] of [drinkt niet] | (3') Jan rookt en drinkt niet |
| (4) dat Jan [niet rookt] of [niet drinkt] | (4') dat Jan niet rookt en drinkt |

Coordination must be a single intonational phrase. An intonational break after “rookt” in, e.g., *Jan rookt, en drinkt niet* (“J. smokes, but doesn't drink”) or *Dat Jan niet rookt, en drinkt!* (“That J. doesn't smoke, but drinks!”) yields different readings, generating structures for which de Morgan's laws are not operative.

Furthermore, NEG-raising (5), Remnant Movement (6), and Raising to SPEC-INFL (7) don't permit RNR/LNR but nevertheless show propositional equivalence of “P en Q” and “P of Q,” *viz.* “neither P nor Q.”

- | |
|---|
| (5) Ik denk niet [dat hij wijn drinkt en/of sigaren rookt] |
| (6) Wijn en/of bier drinken zal Jan niet |
| (7) Dat Jan en/of Piet nog niet gekomen zijn! |

These empirical facts cannot be explained by externalization operations like RNR/LNR but are consistent with the subset condition on the learnability of relative scope of negation and conjunction elements.

[Glosses/translations: (1) J. smokes ~~not~~ and drinks not; (1') J. smokes or drinks not; (2) that J. not smokes and ~~not~~ drinks; (2') that J. not smokes or drinks; (3) J. smokes ~~not~~ or drinks not; (3') J. smokes and drinks not; (4) that J. not smokes or ~~not~~ drinks; (4') that J. not smokes and drinks; (5) "I don't think that he drinks wine and/or smokes cigars"; (6) "Drink wine and/or beer John will not"; (7) "That John and/or Peter have not arrived yet!"]

References

- Barwise, J., & Cooper, R. 1981. Generalized quantifiers and natural language. *Linguistics and Philosophy* 4:159–219.
- Berwick, R.C. 1985. *The Acquisition of Syntactic Knowledge*. Cambridge, MA: MIT Press.
- Bowler, M. 2014. Conjunction and disjunction in a language without 'and.' *Proc. of SALT* 24:137–155.
- Chomsky, N. 1965. *Aspects of the Theory of Syntax*. Cambridge, MA: MIT Press.
- Crain, S. 2008. The interpretation of disjunction in universal grammar. *Lang Speech* 51(1&2):151–169. doi: 10.1177/00238309080510010901
- Crain, S., & R. Thornton. 2011. Syntax acquisition. *WIREs Cogn Sci* 2011. doi: 10.1002/wcs.1158
- von Fintel, K., & Keenan, E.L. 2016. Determiners, Conservativity, Witnesses. *Journal of Semantics* 35(1): 207–217 <https://doi.org/10.1093/jos/ffx018>
- Horn, L.R. 1972. *On the Semantic Properties of Logical Operators in English*, PhD. Diss. UCLA.
- Horn, L.R. 1989. *A Natural History of Negation*. Chicago: University of Chicago Press.
- Huybregts, M.A.C. 1979. De biologische kern van taal. In: Huybregts, R. & L. des Tombe (eds.), *Verkenningen in Taal*, Utrecht: Instituut A.W. de Groot voor Algemene Taalwetenschap.
- Jaspers, D. 2005. *Operators in the Lexicon. On the Negative Logic of Natural Language*. PhD. Diss., Leiden University.
- Knowlton, T.Z, Pietroski, P., Williams, A., Halberda, J., & Lidz, J. 2020. Determiners are “conservative” because their meanings are not relations: evidence from verification. *Proceedings of SALT* 30: 206–226.
- Murray, S.E. 2017. Cheyenne connectives. In: Monica Macaulay, Margaret Noodin, J. Randolph Valentine Monica Macaulay (eds.), *Papers of the 45th Algonquian Conference*, Michigan State University Press. <https://msupress.org/9781611862249/papers-of-the-forty-fifth-algonquian-conference/>
- Singh, R., Wexler, K., Astle-Rahim, A., Kamawar, D., & Fox, D. 2016. Children interpret disjunction as conjunction: Consequences for theories of implicature and child development. *Natural Language Semantics* 24(4), 305–352. <https://doi.org/10.1007/s11050-016-9126-3>
- Yang, C. 2016. *The Price of Linguistic Productivity*. MIT Press, Cambridge, MA.